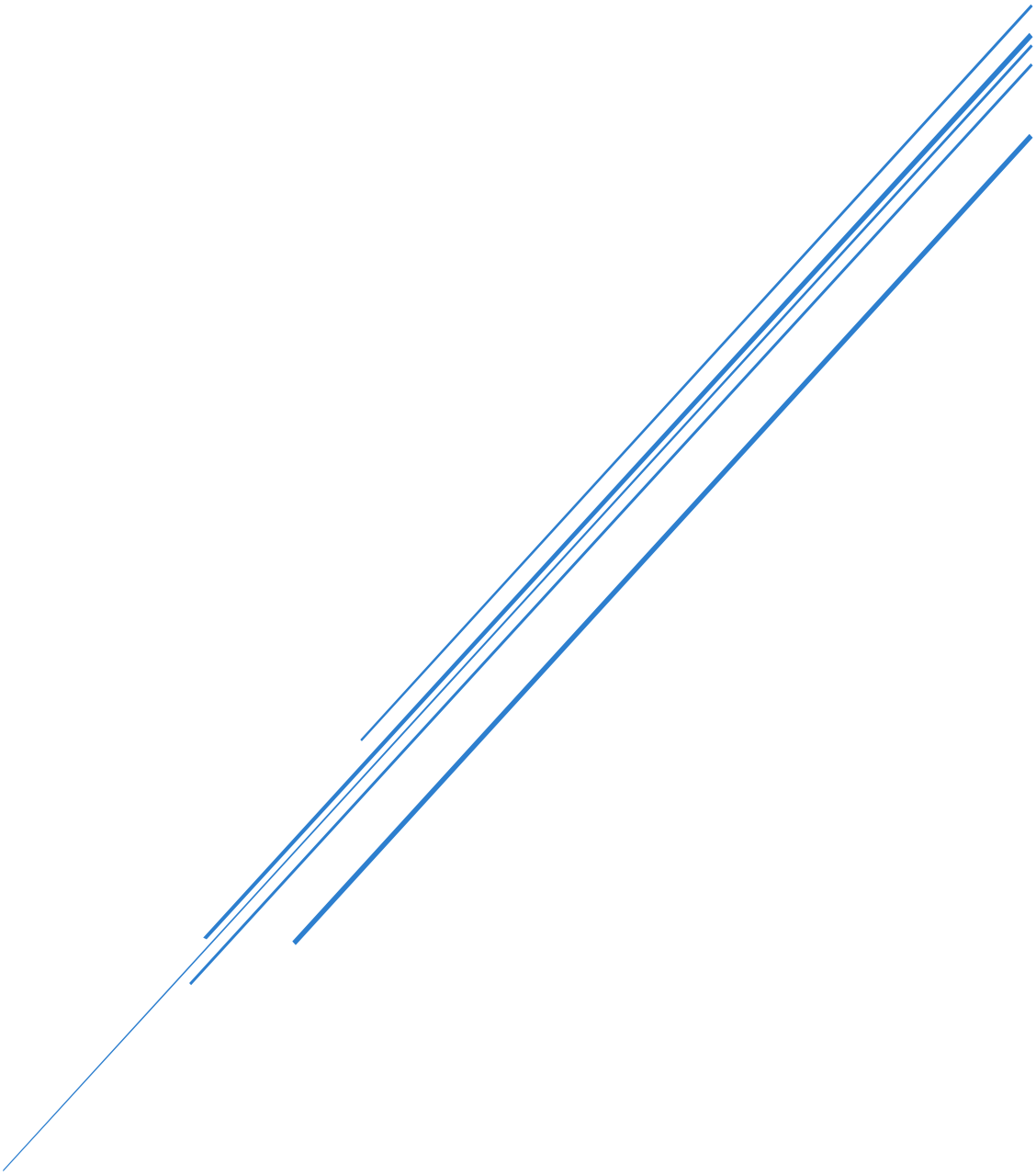
# MOOIFONTEIN – RANCH

Testing Report



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The network was built in stages, each stage was verified before moving on, and the finished network meets the brief. The testing tool used was Cisco Packet Tracer. Each test was documented using the objective, steps, expected result, actual result and pass/fail status. The router configurations and connectivity were tested in Packet Tracer. The *show ip interface brief* command was used to inspect router interface IP addresses and operational status. *Ping* commands were used to verify connectivity between devices. Each stage has its own .pkt file.

This document is the testing evidence for the network designed for Mooifontein Game & Cattle Ranch. Its purpose is to show that the network was built in stages, that each stage was verified before the next was started, and that the finished network meets the requirements in the client brief.

**1. Implementation approach.** The network was built in six stages, each saved as its own Packet Tracer file (Milestone1.pkt to Milestone6.pkt). Building incrementally meant that a fault could be traced to the stage that introduced it, instead of being hidden inside a finished network. The stages were: (1) physical topology, (2) router connections and addressing, (3) switch VLAN configuration, (4) end device configuration, (5) OSPF routing, and (6) ACLs for IoT isolation (CR12).

**2. Testing approach.** Each stage was tested in Packet Tracer using verification commands (such as show ip interface brief, show vlan brief, show ip ospf neighbor and show ip route) and connectivity tests (ping and traceroute). Every test is recorded in the same format: the test objective, the steps and commands used, the expected result, the actual result (with a screenshot), and a pass or fail outcome. Where traffic is meant to be blocked, such as IoT isolation, a failed connection is recorded as a pass, because it confirms the design is enforced.

| **Implementation milestone** | **What was added** | **.pkt file** | **Requirement covered** | **Result** |
|---|---|---|---|---|
| Milestone 1 | .pkt file that includes physical topology with no cabling between routers | Milestone1.pkt | 3-site hub-and-spoke topology | Physical topology only; nothing to test beyond visual confirmation |
| Milestone 2 | .pkt file that includes the connectivity between routers and router addressing | Milestone2.pkt | wireless point-to-point backhaul and IP addressing on routers | |
| Milestone 3 | .pkt file that includes Switch VLAN configuration | Milestone3.pkt | VLAN segmentation per department (Admin, Security, Livestock, | |

| | | | | |
|---|---|---|---|---|
| | | | Veterinary, Game Ops, Rangers, IoT) | |
| Milestone 4 | .pkt which includes end devices configuration | Milestone4.pkt | VLSM addressing from 10.22.0.0/16 with efficient subnet allocation | |
| Milestone 5 | .pkt file which includes the OSPF configuration | Miestone5.pkt | Connecting the routers | Routers are full neighbours and OSPF routes in the routing table after running show ip ospf neighbor and show ip route ospf commands |
| Milestone 6 | .pkt file which includes ACLs for IoT Isolation (CR12) | Milestone6.pkt | IoT isolation (CR12) via dedicated VLANs and router ACLs IoT segments cannot reach administrative subnets | |

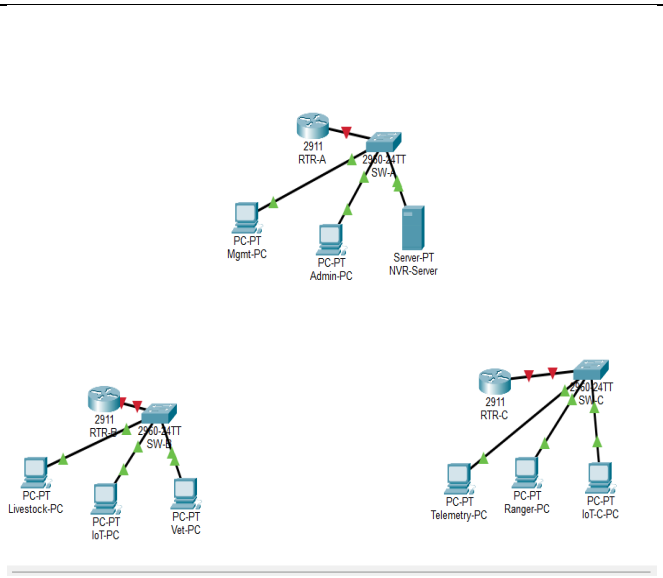
## 3. Addressing plan

| **VLAN** | **Site / Function** | **Network** | **Mask** | **Usable Range** | **Gateway** | **Hosts** | **Classification** |
|---|---|---|---|---|---|---|---|
| 10 | Site A - Administration | 10.22.10.0 | /26 | .1 – .62 | .1 | 62 | High Trust |
| 20 | Site A - Security & Surveillance | 10.22.20.0 | /25 | .1 – .126 | .1 | 126 | Camera/Sensor |
| 30 | Site B - Livestock Management | 10.22.30.0 | /26 | .1 – .62 | .1 | 62 | Operations |
| 40 | Site B - Veterinary | 10.22.40.0 | /27 | .1 – .30 | .1 | 30 | Operations |
| 50 | Site B - IoT Sensors (CR12) | 10.22.50.0 | /24 | .1 – .254 | .1 | 254 | Isolated |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 60 | Site C - Game / Field Ops | 10.22.60.0 | /26 | .1 – .62 | .1 | 62 | Operations |
| 70 | Site C - Ranger Operations | 10.22.70.0 | /27 | .1 – .30 | .1 | 30 | Operations |
| 80 | Site C - IoT Sensors (CR12) | 10.22.80.0 | /24 | .1 – .254 | .1 | 254 | Isolated |
| 99 | Network Management (All Sites) | 10.22.99.0 | /28 | .1 – .14 | .1 | 14 | Infrastructure |
| — | PtP: Site A ↔ B | 10.22.100.0 | /30 | .1 – .2 | — | 2 | Infrastructure |
| — | PtP: Site A ↔ C | 10.22.100.4 | /30 | .5 – .6 | — | 2 | Infrastructure |
| — | PtP: Site B ↔ C (Backup) | 10.22.100.8 | /30 | .9 – .10 | — | 2 | Infrastructure |
| VLAN | Site / Function | Network | Mask | Usable Range | Gateway | Hosts | Classification |
| 10 | Site A - Administration | 10.22.10.0 | /26 | .1 – .62 | .1 | 62 | High Trust |
| 20 | Site A - Security & Surveillance | 10.22.20.0 | /25 | .1 – .126 | .1 | 126 | Camera/Sensor |
| 30 | Site B - Livestock Management | 10.22.30.0 | /26 | .1 – .62 | .1 | 62 | Operations |

## 4. Evidence

| **Implementation Milestone** | **Content** | **Evidence** |
|---|---|---|

| | | |
|---|---|---|
| Milestone 1 | Physical topology (with no cabling between routers) |  |
| Milestone 2 | IP addressing on routers | RTR-A>show ip interface brief<br>Interface IP-Address OK? Method Status Protocol<br>GigabitEthernet0/0 unassigned YES manual up up<br>GigabitEthernet0/0.10 10.22.10.1 YES manual up up<br>GigabitEthernet0/0.20 10.22.20.1 YES manual up up<br>GigabitEthernet0/0.99 10.22.99.1 YES manual up up<br>GigabitEthernet0/1 unassigned YES DHCP up up<br>GigabitEthernet0/2 unassigned YES unset administratively down down<br>Serial0/3/0 10.22.100.1 YES manual up up<br>Serial0/3/1 10.22.100.5 YES manual up up<br>Loopback0 10.22.255.1 YES manual up up<br>Vlan1 unassigned YES unset administratively down down<br>RTR-A><br><br>RTR-B>show ip interface brief<br>Interface IP-Address OK? Method Status Protocol<br>GigabitEthernet0/0 unassigned YES unset up up<br>GigabitEthernet0/0.30 10.22.30.1 YES manual up up<br>GigabitEthernet0/0.40 10.22.40.1 YES manual up up<br>GigabitEthernet0/0.50 10.22.50.1 YES manual up up<br>GigabitEthernet0/1 unassigned YES unset administratively down down<br>GigabitEthernet0/2 unassigned YES unset administratively down down<br>Serial0/3/0 10.22.100.2 YES manual up up<br>Serial0/3/1 10.22.100.9 YES manual down down<br>Loopback0 10.22.255.2 YES manual up up<br>Vlan1 unassigned YES unset administratively down down<br>RTR-B><br><br>RTR-C>show ip interface brief<br>Interface IP-Address OK? Method Status Protocol<br>GigabitEthernet0/0 unassigned YES unset up up<br>GigabitEthernet0/0.60 10.22.60.1 YES manual up up<br>GigabitEthernet0/0.70 10.22.70.1 YES manual up up<br>GigabitEthernet0/0.80 10.22.80.1 YES manual up up<br>GigabitEthernet0/1 unassigned YES unset administratively down down<br>GigabitEthernet0/2 unassigned YES unset administratively down down<br>Serial0/3/0 unassigned YES unset administratively down down<br>Serial0/3/1 10.22.100.6 YES manual up up<br>Loopback0 10.22.255.3 YES manual up up<br>Vlan1 unassigned YES unset administratively down down |

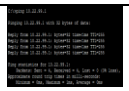
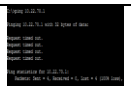
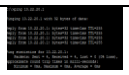
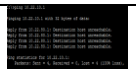
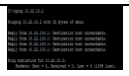
| | | |
|---|---|---|
| Milestone 3 | Switch VLAN configuration | SW-A>show vlan brief<br><br>VLAN Name Status Ports<br>---- -------------------------------- --------- -------------------------------<br>1 default active Fa0/5, Fa0/6, Fa0/7, Fa0/8<br>Fa0/9, Fa0/10, Fa0/11, Fa0/12<br>Fa0/13, Fa0/14, Fa0/15, Fa0/16<br>Fa0/17, Fa0/18, Fa0/19, Fa0/20<br>Fa0/21, Fa0/22, Fa0/23, Fa0/24<br>Gig0/1, Gig0/2<br>10 Administration active Fa0/2<br>20 Security active Fa0/3<br>99 Management active Fa0/4<br>1002 fddi-default active<br>1003 token-ring-default active<br>1004 fddinet-default active<br>1005 trnet-default active<br><br>SW-B>show vlan brief<br><br>VLAN Name Status Ports<br>---- -------------------------------- --------- -------------------------------<br>1 default active Fa0/5, Fa0/6, Fa0/7, Fa0/8<br>Fa0/9, Fa0/10, Fa0/11, Fa0/12<br>Fa0/13, Fa0/14, Fa0/15, Fa0/16<br>Fa0/17, Fa0/18, Fa0/19, Fa0/20<br>Fa0/21, Fa0/22, Fa0/23, Fa0/24<br>Gig0/1, Gig0/2<br>30 Livestock active Fa0/2<br>40 Veterinary active Fa0/3<br>50 IoT-Sensors active Fa0/4<br>1002 fddi-default active<br>1003 token-ring-default active<br>1004 fddinet-default active<br>1005 trnet-default active<br><br>SW-C>show vlan brief<br><br>VLAN Name Status Ports<br>---- -------------------------------- --------- -------------------------------<br>1 default active Fa0/5, Fa0/6, Fa0/7, Fa0/8<br>Fa0/9, Fa0/10, Fa0/11, Fa0/12<br>Fa0/13, Fa0/14, Fa0/15, Fa0/16<br>Fa0/17, Fa0/18, Fa0/19, Fa0/20<br>Fa0/21, Fa0/22, Fa0/23, Fa0/24<br>Gig0/1, Gig0/2<br>60 GameOps active Fa0/2<br>70 Rangers active Fa0/3<br>80 IoT-Sensors active Fa0/4<br>1002 fddi-default active<br>1003 token-ring-default active<br>1004 fddinet-default active<br>1005 trnet-default active |
| Milestone 4 | Configure end devices | Pinging 10.22.40.2 with 32 bytes of data:<br><br>Reply from 10.22.40.2: bytes=32 time<1ms TTL=128<br>Reply from 10.22.40.2: bytes=32 time<1ms TTL=128<br>Reply from 10.22.40.2: bytes=32 time=4ms TTL=128<br>Reply from 10.22.40.2: bytes=32 time=4ms TTL=128<br><br>Ping statistics for 10.22.40.2:<br>Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),<br>Approximate round trip times in milli-seconds:<br>Minimum = 0ms, Maximum = 4ms, Average = 2ms<br><br>C:\>ping 10.22.10.1<br><br>Pinging 10.22.10.1 with 32 bytes of data:<br><br>Reply from 10.22.10.1: bytes=32 time=28ms TTL=255<br>Reply from 10.22.10.1: bytes=32 time<1ms TTL=255<br>Reply from 10.22.10.1: bytes=32 time=1ms TTL=255<br>Reply from 10.22.10.1: bytes=32 time<1ms TTL=255<br><br>Ping statistics for 10.22.10.1:<br>Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),<br>Approximate round trip times in milli-seconds:<br>Minimum = 0ms, Maximum = 28ms, Average = 7ms |

| | | |
|---|---|---|
| | | **Pinging 10.22.80.2 with 32 bytes of data:**<br><br>**Reply from 10.22.80.2: bytes=32 time<1ms TTL=128**<br>**Reply from 10.22.80.2: bytes=32 time=4ms TTL=128**<br>**Reply from 10.22.80.2: bytes=32 time<1ms TTL=128**<br>**Reply from 10.22.80.2: bytes=32 time=6ms TTL=128**<br><br>**Ping statistics for 10.22.80.2:**<br>**Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),**<br>**Approximate round trip times in milli-seconds:**<br>**Minimum = 0ms, Maximum = 6ms, Average = 2ms** |
| Milestones 5 | OSPF configuration | RTR-C>enable<br>RTR-C#show ip ospf neighbor<br><br>Neighbor ID Pri State Dead Time Address Interface<br>10.22.255.1 0 FULL/ - 00:00:34 10.22.100.5 Serial0/3/1<br>RTR-C#show ip route ospf<br>10.0.0.0/8 is variably subnetted, 16 subnets, 7 masks<br>O 10.22.10.0 [110/65] via 10.22.100.5, 00:00:43, Serial0/3/1<br>O 10.22.20.0 [110/65] via 10.22.100.5, 00:00:43, Serial0/3/1<br>O 10.22.30.0 [110/129] via 10.22.100.5, 00:00:43, Serial0/3/1<br>O 10.22.40.0 [110/129] via 10.22.100.5, 00:00:43, Serial0/3/1<br>O 10.22.50.0 [110/129] via 10.22.100.5, 00:00:43, Serial0/3/1<br>O 10.22.99.0 [110/65] via 10.22.100.5, 00:00:43, Serial0/3/1<br>O 10.22.100.0 [110/128] via 10.22.100.5, 00:00:43, Serial0/3/1<br><br>RTR-B>enable<br>RTR-B#show ip ospf neighbor<br><br>Neighbor ID Pri State Dead Time Address Interface<br>10.22.255.1 0 FULL/ - 00:00:38 10.22.100.1 Serial0/3/0<br>RTR-B#show ip route ospf<br>10.0.0.0/8 is variably subnetted, 16 subnets, 7 masks<br>O 10.22.10.0 [110/65] via 10.22.100.1, 00:01:16, Serial0/3/0<br>O 10.22.20.0 [110/65] via 10.22.100.1, 00:01:16, Serial0/3/0<br>O 10.22.60.0 [110/129] via 10.22.100.1, 00:01:06, Serial0/3/0<br>O 10.22.70.0 [110/129] via 10.22.100.1, 00:01:06, Serial0/3/0<br>O 10.22.80.0 [110/129] via 10.22.100.1, 00:01:06, Serial0/3/0<br>O 10.22.99.0 [110/65] via 10.22.100.1, 00:01:16, Serial0/3/0<br>O 10.22.100.4 [110/128] via 10.22.100.1, 00:01:16, Serial0/3/0<br><br>RTR-A>enable<br>RTR-A#show ip ospf neighbor<br><br>Neighbor ID Pri State Dead Time Address Interface<br>10.22.255.3 0 FULL/ - 00:00:31 10.22.100.6 Serial0/3/1<br>10.22.255.2 0 FULL/ - 00:00:31 10.22.100.2 Serial0/3/0<br>RTR-A#show ip route ospf<br>10.0.0.0/8 is variably subnetted, 17 subnets, 7 masks<br>O 10.22.30.0 [110/65] via 10.22.100.2, 00:00:28, Serial0/3/0<br>O 10.22.40.0 [110/65] via 10.22.100.2, 00:00:28, Serial0/3/0<br>O 10.22.50.0 [110/65] via 10.22.100.2, 00:00:28, Serial0/3/0<br>O 10.22.60.0 [110/65] via 10.22.100.6, 00:00:28, Serial0/3/1<br>O 10.22.70.0 [110/65] via 10.22.100.6, 00:00:28, Serial0/3/1<br>O 10.22.80.0 [110/65] via 10.22.100.6, 00:00:28, Serial0/3/1 |
| Milestones 6 | ACLs for IoT Configuration (CR12) | Ping from IoT-PC to Admin-PC<br>**C:\>ping 10.22.10.2**<br><br>**Pinging 10.22.10.2 with 32 bytes of data:**<br><br>**Request timed out.**<br>**Request timed out.**<br>**Request timed out.**<br>**Request timed out.**<br><br>**Ping statistics for 10.22.10.2:**<br>**Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),**<br><br>Ping from IoT-C-PC to Vet-PC |

| | | |
|---|---|---|
| | | C:\>ping 10.22.40.2<br><br>Pinging 10.22.40.2 with 32 bytes of data:<br><br>Request timed out.<br>Request timed out.<br>Request timed out.<br>Request timed out.<br><br>Ping statistics for 10.22.40.2:<br>Packets: Sent = 4, Received = 0, Lost = 4 (100% loss), |

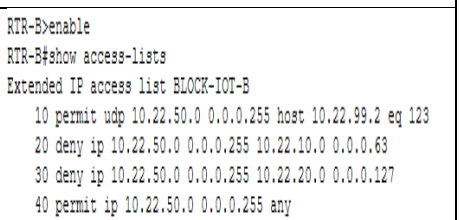
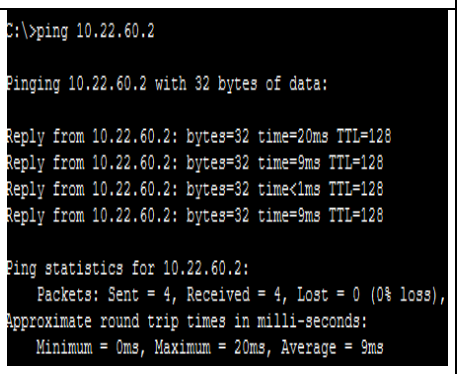
## 5. End-to-end check on the final network

All tests were run on the final implementation milestone (Milestone 6). These tests cover each requirement on the brief as well as ensuring that the ACLs did not break earlier stages.

| Test-ID | Source | Destination | Purpose | Expected | Evidence | Result |
|---|---|---|---|---|---|---|
| E1 | Livestock-PC | Vet-PC | Inter-VLAN routing Site B | Success |  | Pass |
| E2 | Admin-PC | Mgmt-PC | Inter-VLAN routing Site A | Success |  | Pass |
| E3 | Ranger-PC | Telemetry-PC | Inter-VLAN routing Site C | Success |  | Fail |
| E4 | Admin-PC | Livestock-PC | Site A to B over OSPF | Success |  | Pass |
| E5 | Admin-PC | Ranger-PC | Site A to C over OSPF | Success |  | Pass |
| E6 | Vet-PC | Telemetry-PC | Site B to C over OSPF | Success |  | Fail |
| E7 | Ranger-PC | NVR-Server | Site C to A server access | Success |  | Fail |
| E8 | IoT-PC | Admin-PC | IoT blocked from Admin | Blocked |  | Pass |
| E9 | IoT-C-PC | Admin-PC | IoT blocked from Site C | Blocked |  | Pass |
| E10 | IoT-PC | Allowed destination | Shows the ACL only blocks what it should | Success |  | Pass |
| E11 | Ranger-PC | Admin-PC | - | Blocked |  | Pass |
| E12 | Livestock-PC | Admin-PC | - | Blocked |  | Pass |

## 6. Issues found and fixed

Testing exposed the following issues. Each was recorded as a failed test, fixed, and re-tested; the evidence for each is included in the last column.

| Test-ID | Symptom | Tests Affected | Root cause | Fix | Retest Evidence |
|---|---|---|---|---|---|
| A1 | Nothing restricted non-IoT VLANs from reaching VLAN 10, where customer records are held | E11 & E12 | The only ACL in place covered IoT (CR12), so the confidentiality constraint was only partly met | Added ACL PROTECT-ADMIN on RTR-A, outbound on Gi0/0.10 |  |
| A2 | End devices (PCs and server) had IP addresses, masks or gateways that didn't match the addressing plan | E3, E6 & E7 | Devices were addressed before the plan was finalised | Fixed addressing according to the addressing plan |  |

## 7. Requirements traceability and conclusion

| ID | Requirement | How the design meets it | Evidence | Status |
|---|---|---|---|---|
| R1 | Use the assigned address block 10.22.0.0/16 | VLSM subnets per VLAN, /30 links, /32 loopbacks | Addressing plan; show ip interface brief on RTR-A/B/C (Stage 2) | Pass |
| R2 | Connect the three sites | Serial links A–B and A–C, backup B–C | Implementation milestone 2; E4, E5, E6 | Pass |
| R3 | Basic OSPF (challenge) | Single-area OSPF, Area 0 | show ip ospf neighbor, show | Pass |

| | | | | |
|---|---|---|---|---|
| | | | ip route ospf ;E4–E6 | |
| R4 | Department segmentation | One VLAN per department, trunks to routers | show vlan brief, show interfaces trunk; E1–E3 | Pass |
| R5 | Working end devices and network services | Static addressing per VLAN, NVR server at Site A | ipconfig audit table; E1–E7 | Pass |
| R6 | CR12: IoT sensors on an isolated segment | VLANs 50 and 80 with ACLs BLOCK-IOT-B/C | ACL counters; E8, E9, E10 | Pass |
| R7 | Constraint: customer records confidential, access controlled | ACL PROTECT-ADMIN on RTR-A protecting VLAN 10 | E11, E12, authorised-access test, ACL counters | Pass |
| R8 | Working, testable Packet Tracer implementation | One .pkt file per stage, final file tested end to end | Milestone1–6 (plus 7, if you save the records ACL separately); | Pass |

## Conclusion

The network for Mooifontein Game & Cattle Ranch was built in six stages and verified at each one. The final implementation connects three sites using OSPF, segments each site into VLANs from the assigned 10.22.0.0/16 block, isolates the IoT sensor networks as required by CR12, and restricts access to the Administration network where customer records are held.

Testing found 3 issues, mainly end-device addressing that did not match the plan and a gap in the protection of customer records. Each was recorded, fixed and re-tested. In the final end-to-end run, tests passed, including the blocked-traffic tests, which were confirmed against ACL match counters so that a block could be attributed to the ACL and not to a misconfiguration.

ACLs are stateless, so hosts outside Administration cannot be pinged from Administration without explicit reply permits / the B–C link is a redundant path, and OSPF costs were not tuned. The design is considered to meet the requirements of the brief.